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Optimization of NiO Nanoparticles Synthesis by Coprecipitation for Advanced Applications

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1-Abstract:

Nickel oxide nanoparticles (NPs) have garnered significant interest due to their unique physicochemical properties, which make them suitable for various advanced technological applications. NiO NPs can be synthesized using various methods such as sol-gel, coprecipitation , hydrothermal synthesis, and thermal decomposition, each offering distinct advantages and influencing the properties of the NPs differently. Thus, optimizing synthesis methods to produce NiO NPs with tailored properties remains a critical challenge in nanotechnology. The present work focuses on synthesizing NiO nanoparticles via a precipitation method .

2-Synthesis of NiO Nanoparticles by Chemical Precipitation Method

.Methodology :

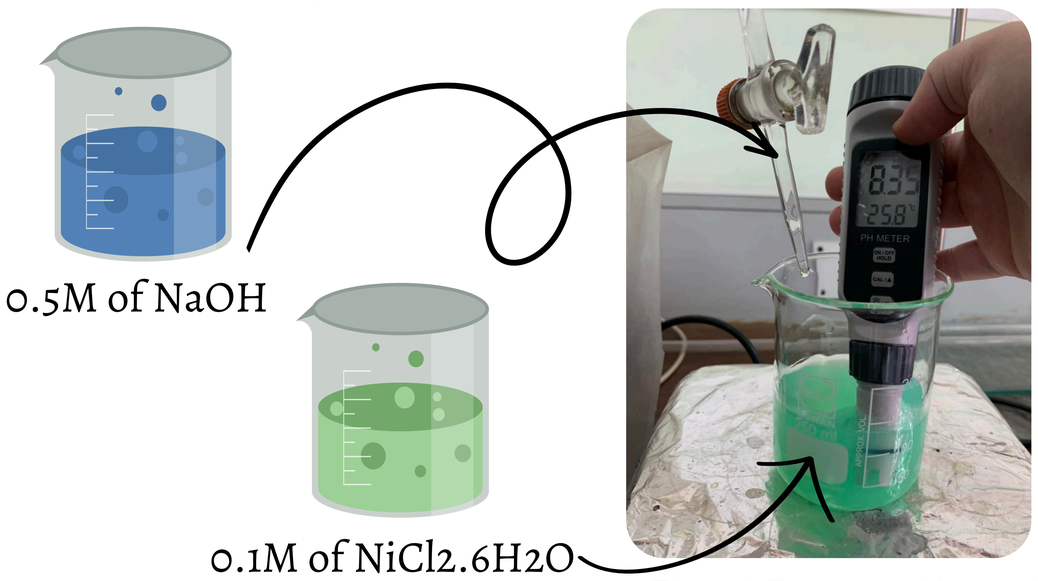


Figure 1 : Dropwise addition of NaOH to adjust the pH

1. Sodium hydroxide (NaOH) solution was added dropwise to the mixture under continuous stirring (400 rpm) until a pH of 9 was reached.



Figure 2 : Water bath

2. Stirring the mixture for 30 min at 40 °C in a water bath (figure2).



Figure 3 : wash by centrifugation.

3. The resulting light green suspension was subjected to multiple centrifugation cycles and washed three times with distilled water (figure3) .

4. The resulting green gel was oven-dried at 60°C for 12 hours.

5. The obtained nickel hydroxide ($Ni(OH)_2$) powder was subsequently calcined at 450°C for 3 hours in a muffle furnace.

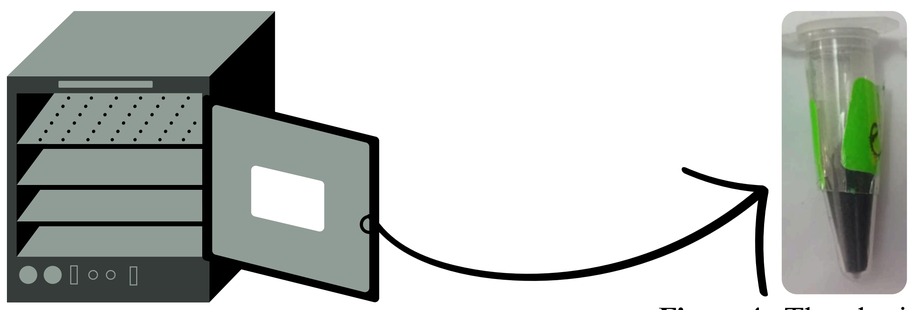


Figure 4 : The obtained Sample.

3-Characterization of NiO Nanoparticles :

.Results and Discussion :

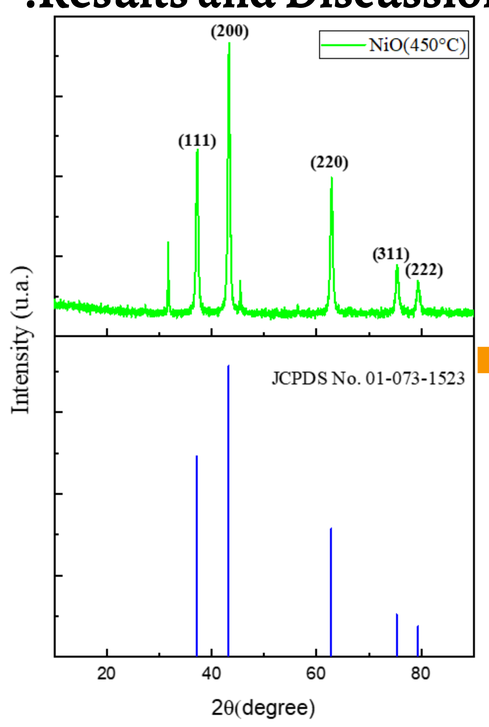


Figure 5 :X-ray diffractogram of the NiO NPs .

3-a- XRD Analysis :

- Peak indexing was performed using X'PertHighScore© software.
- Consistent with the standard NiO of **[JCPDS Card No.01-073-1523]** .
- NiO NPs exhibited a face-centered cubic (FCC) structure .
- The average crystallite size was estimated using the Debye-Scherrer equation is **22.7 nm** .

3-b- Raman Analysis :

- The Raman spectra confirmed the presence of **Ni-O** bonds.

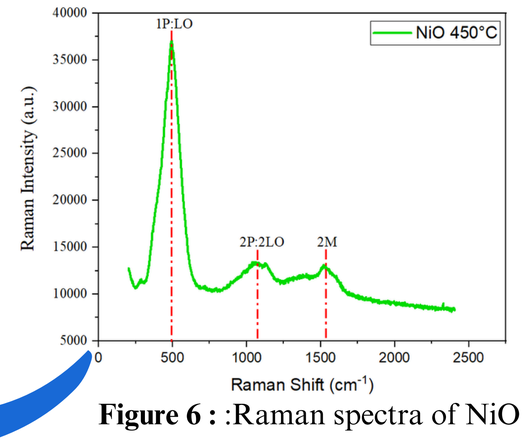


Figure 6 : : Raman spectra of NiO NPs annealed at 450°C.

3-c- SEM Analysis :

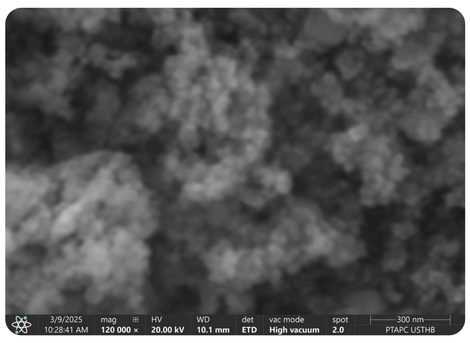
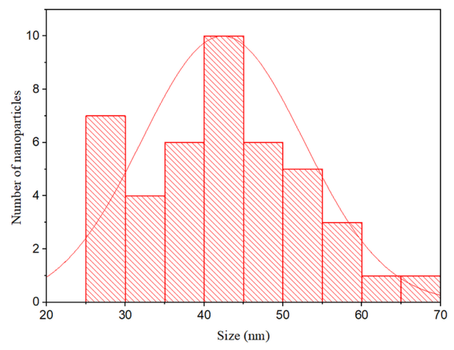


Figure 7 : SEM image of NiO NPs

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- Spherical shape.
- The size of the NPs is smaller than **45 nm .**

4- conclusion:

- The synthesis of NiO nanoparticles (NPs) using the coprecipitation method at a calcination temperature of 450°C is an effective approach to produce nanoparticles with very small size and high purity.
- The results demonstrate that calcination at 450°C facilitates controlled growth of nickel oxide (NiO) nanoparticles while mitigating the risk of agglomeration .these results hold promising potential for integrating these nanoparticles in the development of advanced technological applications such as Sensors , Batteries and spintronic devices .